Académie militaire de Saint-Cyr Coëtquidan

Benchmarking of a LoRa Mesh Network

for Tactical Communications

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Abstract

This report presents the design and experimental evaluation of a wireless mesh network based on the open-source Meshtastic project. The goal is to assess LoRa (Long Range) modulation and the Meshtastic protocol for decentralised, low-power, long-range communications that remain functional without conventional infrastructures (military radios, hertzian antennas, costly systems). Such networks suit low-bandwidth traffic and crisis scenarios with jamming, where robust, autonomous links are vital.

After outlining the LoRa fundamentals and the Meshtastic protocol, this project presents a three step experimental process that aims at assessing the performance and usability of a LoRa mesh network to support tactical communications: blue force tracking, text messaging, as well as sensor data monitoring. The first step consists in measuring LoRa coverage capabilities in different environments using Meshtastic to determine the optimal radio settings and node positioning. Then, a Meshtastic network has been deployed following the previous conclusions in order to evaluate its performance. Eventually, a tactical communication scenario was played to assess the relevance of the solution for military usage.

Results showed that LoRa mesh provides multi-kilometer coverage under favorable conditions with acceptable reliability. Indeed, experiments showed that LoRa can cover large areas with high-ground positioned devices, using very little power. The conducted experiments highlighted strengths such as a natural resilience favored by the mesh topology, and high autonomy, especially for dropped sensors that only wake up to transmit relevant pieces of information. They also highlighted limits such as low throughput and fixed radio settings preventing self-adaptive behaviors on the nodes. LoRa mesh networking was successfully used for carrying tactical communications on an area spanning from the *Grande Bosse* forest to the academy classroom facilities.

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1 Introduction

1.1 Motivations

Wireless communications usually depend on centralised infrastructure (cell towers, Wi-Fi APs, satellites). In remote areas, disaster relief situations, or military contexts, where network infrastructures are not available, independent and resilient systems are essential. Mesh networks allow each node to relay traffic and maintain connectivity without a single control point. This kind of communication is usually used to implement tactical awareness on the battlefield. The nodes of a tactical mesh network typically share data such as text (or voice) communications, such as orders or reports, share locations, waypoints or any geo-localised relevant piece of information. They can also be used to collect or track sensors dropped on the field to gather intelligence, e.g., on enemy's activities.

Since the start of the Russian–Ukrainian war, the demand for low-tech, reliable, and affordable equipment has grown. Armed forces must balance cost and capability, driving interest in robust, low-cost off-the-shelf solutions. LoRa technology, despite its low data rate, provides properties such as low power consumption, long-range, and discretion, that might be relevant for a military usage. Several works published in the academic and military literature have already studied LoRa technology over LoRaWAN for military IoT [7, 11, 13] or for mesh networking [5, 15]. Meshtastic [9] leverages LoRa to create mesh networks of easily accessible and affordable radio nodes where each device sends, receives, and relays messages, enabling long-range communication for outdoor activities, rescue missions or even military operations.

This study therefore aims at evaluating the performance and the relevance of LoRa and Meshtastic protocol when used as the networking layer to support tactical communications.

1.2 Objectives

The goal of this project is to validate the relevance of LoRa technology in a tactical context; accordingly, we chose to evaluate Meshtastic, emerging, open-source solution that enables the rapid deployment of scalable LoRa mesh networks using inexpensive hardware.
We have chosen to organise our study in three phases :

- Range tests (peer-to-peer) in several conditions and environments. The results of these tests should help us to determine the optimal LoRa settings and node positioning in the mesh for our further experiments.
- Deployment of the mesh network on a wide area and performance measurements. This phase should allow us to evaluate the performance of the mesh network : reliability (packet delivery ratio), network load (number of exchanged messages), latency, and extended coverage capabilities using multi-hop communications.
- Use of this network in a tactical scenario : this last phase should enable us to evaluate the feasibility of such networks in the field. Eventually, it could mean sharing tactical awareness (Blue Force Tracking), exchanging text messages (orders and reports) and gathering intelligence via sensors already laid on the field.

Through this dual approach—technical implementation and operational evaluation—the project seeks to assess the potential of Meshtastic as a reliable, low-cost, low-power communication solution for tactical or crisis-response scenarios where autonomy and resilience are critical.

In the end, we would like to share our results with the scientific community interested in military communications in the ICMCIS conference organized by NATO's STO (Science and Technology Organization).

1.3 Project Management

Our first priority was to define clear project boundaries so we could plan effectively—deciding what to do and where to stop. The topic is rich and could be explored far beyond our scope, which is precisely why setting limits was essential.

Figure 1 presents the chronological organisation that we followed. At first, we needed to get along with Meshtastic and adapt it to our needs. Indeed, the firmware itself does not allow data to be exploited, the reason for which a functional logger needed to be developed, along with an optimisation of the memory use and the forwarding of packets within the network.

For team management and collaborative work, we used several complementary tools: a Microsoft Teams chat room for exchanging messages and files, as well as for keeping track of project progress; Git for tracking firmware changes via patches and version management; a Gantt chart for planning and time management; and latex on Overleaf for collaborative writing and proofreading.

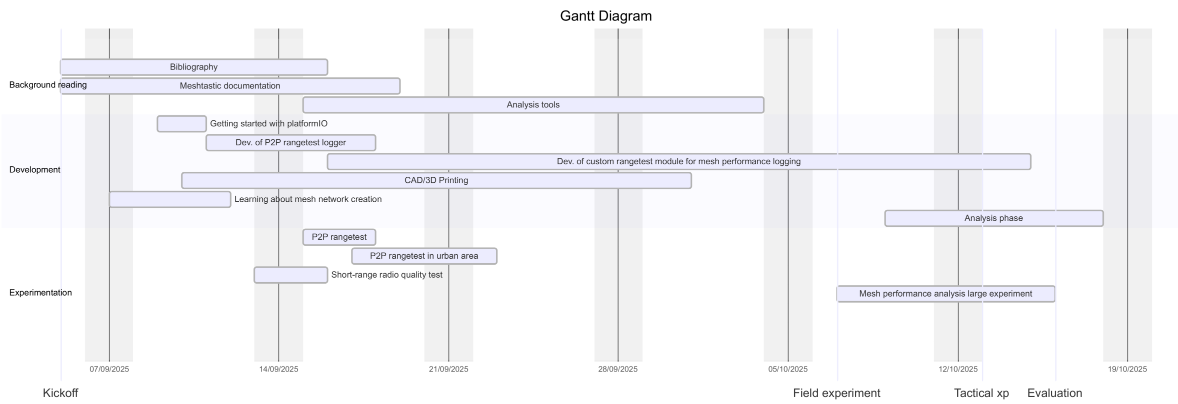


Figure 1: Project GANTT diagram

Name	Role	Tasks
2LT LIGIER	Team leader	Firmware development Data analysis Report overview
2LT DEMADE	Experiment manager	Planning Experience architect Data analysis
2LT DE FAUTEREAU-VASSEL	Principal writer	Information gathering Translation Redaction overview

Table 1: Role distribution

2 LoRa and Meshtastic overview

This section gives an overview of LoRa physical layer and its properties. It presents which MAC layer protocols can be used on LoRa PHY layer, including LoRaWAN for IoT uses, or ad hoc protocols for building LoRa mesh networks. Then this section presents Meshtastic, a LoRa mesh protocol, that has been gaining more and more attention.

2.1 LoRa and its protocols

LoRa is a long range and low power radio technology based on chirp spread spectrum modulation which operates on ISM frequency bands like 868MHz or 433MHz in Europe. The LoRa modulation allows good performances under low SNR and RSSI doubled with a low energy consumption, which results in stealthy communications. In the beginning, LoRa was developed to be used in IoT mainly relying on a protocol layer called LoRaWAN.

A 'chirp', as shown on Figure 2 is a signal whose frequency linearly sweeps a band (upward or downward) during each symbol; LoRa encodes the information by shifting these sweeps in time, which makes detection robust at low SNR. The spreading factor (SF) indicates the number of logical bits carried by a CSS symbol via $2^S F$ 'chirps': the higher the SF, the longer the symbol and the greater the receiver sensitivity, improving range and robustness to interference and fading. On the other hand, the data rate drops and the time-on-air and risk of collisions increase. Thus, choosing the SF is a trade-off between range/reliability and data rate/radio load.

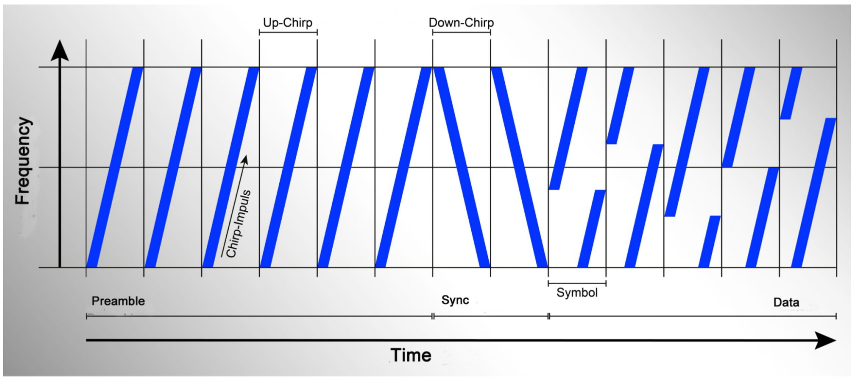


Figure 2: LoRa modulation

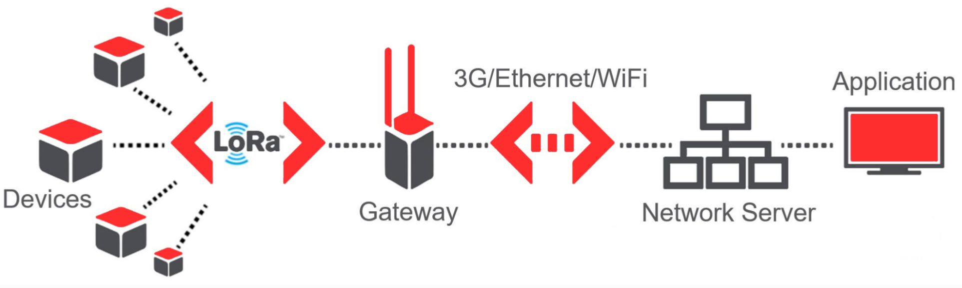


Figure 3: LoRaWAN architecture for IoT

LoRaWAN is the standard MAC layer protocol over LoRa, defined by the LoRa Alliance for IoT networks following a star topology. While LoRa defines the physical layer, LoRaWAN establishes the rules for data exchange, frame formats, timing, and network management. The LoRaWAN architecture uses mainly uplink communications from sensor devices to gateways (see left part on Figure 3), thus enabling scalable, low-power IoT networks that can support thousands of devices with reliable, long-range connectivity and increased lifetime.

LoRa Mesh. In this project we intend to use LoRa technology to support tactical communications through mesh networking.

In a full mesh network, every node maintains a direct link to all others, providing maximum path redundancy and single-hop latency, but at the expense of escalating radio complexity (collisions, duty-cycle pressure) and cost as the network scales. By contrast, a partial mesh connects only a subset of node pairs, using relays to preserve end-to-end connectivity; this curbs channel load while retaining adequate resilience. For LoRa, a partial mesh with a few elevated backbone routers offers a practical balance of range, reliability, and spectral efficiency. In this project on tactical communications, we therefore seek to employ LoRa to build a resilient, energy-efficient mesh.

Although there is no established standard for mesh networking over LoRa, the literature has proposed some solutions [1,2,4,8,10,12], including Meshtastic [3,9,14] a rising community-driven open-source project. In this study we adopt Meshtastic for its software maturity, low hardware cost, and active community support. In principle, our type of network is a complete mesh network with no hierarchy, in order to offer great ease of use and redundancy. It may look like a partial mesh network, but this is only due to the elongation between certain nodes and not to the intrinsic structure of the network.

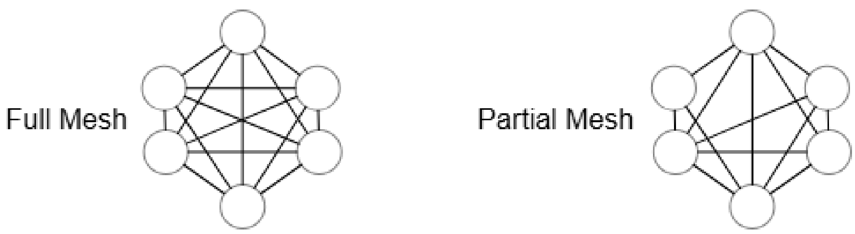


Figure 4: Mesh networking

2.2 Meshtastic firmware and protocol

Meshtastic is an open-source mesh networking project built on LoRa radios, designed to provide resilient communication in environments where infrastructure-based networks such as cellular or the Internet are unavailable. By leveraging inexpensive, low-power hardware, it enables multi-hop text messaging, position sharing, and telemetry across several kilometers, making it suitable for both recreational use and critical off-grid scenarios. Unlike LoRaWAN, which relies on centralised gateways, Meshtastic employs a peer-to-peer mesh architecture where every device can act as both an endpoint and a router.

The forwarding protocol underpinning Meshtastic is deliberately lightweight to suit the constraints of LoRa's low data rate and strict duty-cycle limits. Messages are relayed across the mesh using a priority mechanism: nodes with weaker received signal strength (lower RSSI) are given forwarding precedence, which helps extend coverage by favoring retransmissions from more distant peers as shown on Figure 6.

To avoid unnecessary duplication and channel congestion, acknowledgements are implicit and retransmissions are bounded. Communication compatibility further depends on channel alignment: all participating devices must share the same radio configuration—carrier frequency, bandwidth, and spreading factor—ensuring interoperability while also segmenting independent meshes. This design balances simplicity, robustness, and scalability within the severe bandwidth limitations of the underlying physical layer (LoRa 433MHz or 868MHz for EU region). Meshtastic MAC layer also relies on specific packet headers to implement routing headers in addition to application-specific payloads that target application ports (*portnum*, e.g., $TEXT, ROUTING, POSITION, ...$). The nodes communicate on the same physical and logical channels, and follow a listen before talk approach based on channel activity detection (CAD), allowing carrier-sense multiple access with collision avoidance (CSMA/CA).

- Channels: a main (primary) channel for general discussion, sharing positions, etc. This channel is not the only one that can be used; other channels (secondary channels) can also be used, typically to distinguish between the channel used for issuing orders and another for logistics, etc.
- The triplet: each channel is defined by a triplet (radio settings: LoRa physical parameters, channel name: unique text identifier to define the channel, PSK: Pre-shared key, allows packet encryption and decryption, it is a 256-bit key).
- Encryption: Communications are encrypted using AES-256-CTR, a stream cipher suitable for low LoRa data rates.

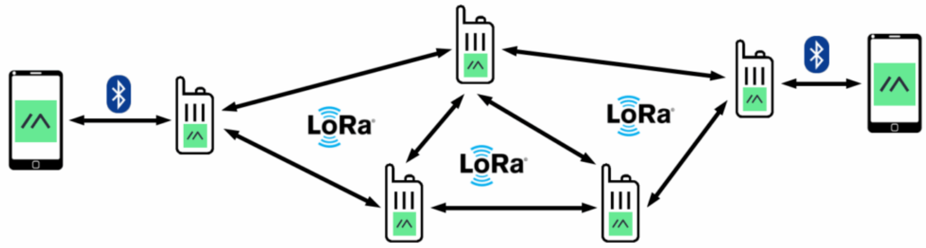


Figure 5: Meshtastic network idea

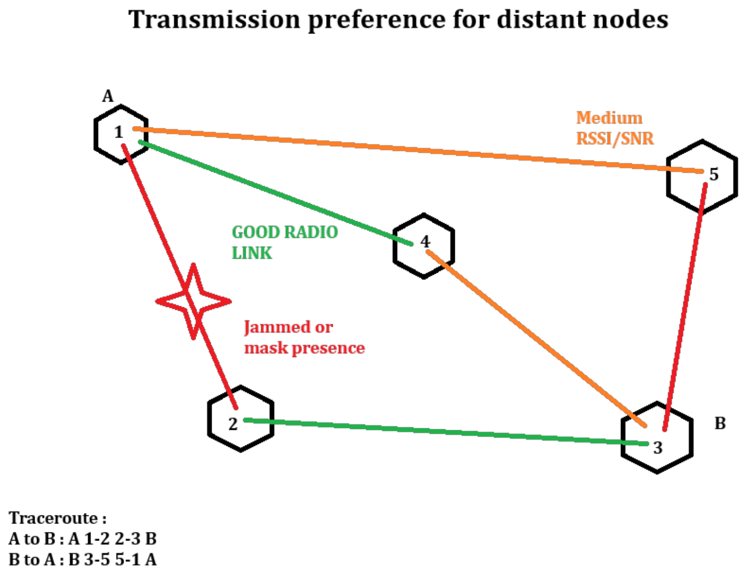


Figure 6: Meshtastic forwarding protocol

- Limitation: If a node in the network is compromised, this jeopardises the security of transmissions because each node contains the channel keys. Thus, seeing a compromised node would be tantamount to deliberately leaving the network open to eavesdropping.

The Meshtastic ecosystem is broad and mature, comprising cross-platform clients (a web UI for configuration and monitoring, plus a Python library/CLI for automation and analysis), add-ons such as Paxcounter for Wi-Fi/BLE people counting, and third-party integrations like an ATAK plug-in for real-time tactical workflows. Sustained by a highly active community (forums, GitHub, Discord, Meshtastic workshops organized by universities) that documents, tests, and iterates rapidly, it delivers frequent fixes and features, wide hardware support, and shared, field-proven radio profiles. Altogether, this accelerates operational deployment, strengthens experimental reproducibility, and fosters research and development that would be hard to achieve in isolation.

3 Experimental protocol

We conducted several experiments to evaluate both radio performance and, more broadly, the network's ability to efficiently transmit additional information while managing Meshtastic's usual traffic, including packet routing, positions, and telemetry. This section describes the hardware used, software adaptations, and the experimental protocol following a three-phase approach.

3.1 Experimental resources

To perform the measurements, ESP32 microcontroller development boards integrating LoRa transceivers were used with Meshtastic firmware installed. This section describes the hardware and software developments required to conduct the experiments.

3.1.1 Hardware

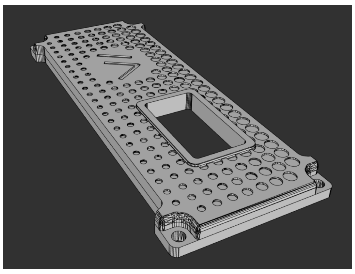
The equipment provided for this project was composed of *Heltec* and *T-Beam* devices whose properties are described in Table 2. In addition we used two 4.4 dB gain antennas.

Table 2: ESP32 development boards

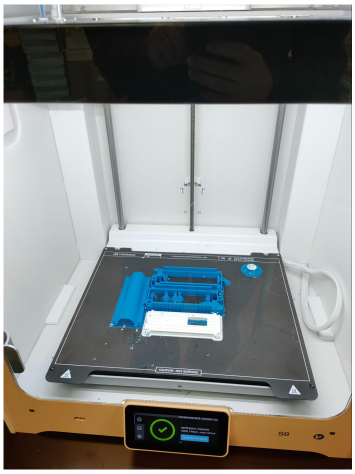
Model	**MCU**	**LoRa**	**Wifi**	**BLE**	**GNSS**	**Display**	**Battery**	**Quantity**
HELTEC V3	ESP32-S3	868MHz (SX1262)	0	0	x	LCD screen	LiPo 3000mAh	8
Lyligo T-BEAM Suprême	ESP32	433/868MHz (SX1262)	x	x	x	Li-ion 2900mAh	OLED	3+4

The Heltec WiFi LoRa 32 (V3) is an all-in-one IoT development board designed for long-range, low-power applications. It combines an ESP32-S3 MCU, an SX1262 LoRa modem and a small OLED screen in a compact format, powered by USB-C or a Li-ion battery. It is actively supported by the Meshtastic community. It is relatively inexpensive, costing around €20 without any economies of scale. Development on this platform is relatively easy, as the board can be flashed with firmware as soon as it is powered up, without the need for a bootloader mode. The T-Beam S3 Core Supreme is a field-oriented LoRa/ESP32-S3 development platform designed for Meshtastic and GNSS tracking. It combines an ESP32-S3 MCU, an SX1262 LoRa modem, a GNSS module (here the L76K version), a dedicated PMU, integrated sensors and, depending on the batch, a small OLED screen.

qIn order to protect equipment in the field, we designed dedicated protective cases. Several community options (on Printables, Thingiverse, etc.) were considered, but we needed a simple, robust, and above all, a rain-resistant design for Heltec V3 modules used as relays. We therefore designed a two-piece enclosure that allows the module, its antenna, and battery to be inserted, and then closes with a cap. This geometry reduces potential points of water ingress and facilitates 3D printing and field maintenance.



(a) 3D Model view

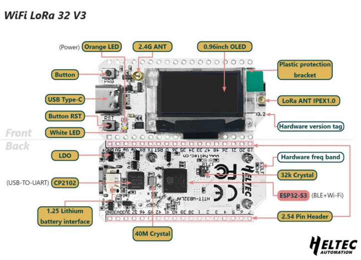


(b) FDM Printed case

(c) Resulting cased T-Beam

Figure 9: T-Beam case design and manufacturing process

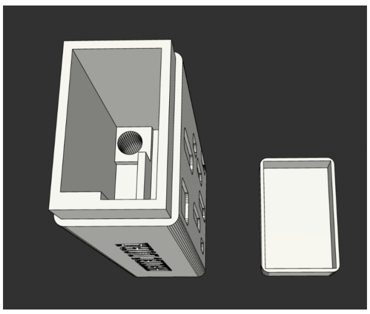
The manufacture of casings using 3D printing is part of a low-tech approach consistent with our objectives of simplicity and cost efficiency. Consumer additive manufacturing processes (FDM/FFF) enable the production of functional, customisable and repairable enclosures without the need for plastics moulding tools, with minimal investment in equipment. In challenging or logistically constrained environments, the ability to design, iterate and manufacture robust enclosures locally is a major operational advantage.



(a) Heltec LoRa v3

(b) Lilygo T-Beam supreme

Figure 7: ESP32-based hardware



(a) 3D Model .3mf

(b) FDM Printed case

Figure 8: Two-piece case enclosure for HELTEC v3 modules printed with inexpensive FDM printer

3.1.2 Software

Custom range test module Meshtastic's native logging capabilities proved insufficient for our purposes. The built-in logger captured some range test data, but only in a peer-to-peer range test scenario. Since the range test data was not easily accessible, we decided to design our own modified module. Furthermore, as the initial module was designed for range test logging, it was incompatible with measuring the performance of a mesh network. We therefore modified the module and selected a useful subset of information per packet. This module is based on version *2.6.11.60ec05e Beta* of the meshtastic firmware. As meshtastic is still undergoing rapid development, we have chosen to select a stable version to ensure the stability of our module.

This custom module is designed to measure the performance of the entire mesh network, recording in a CSV file, on each node, the chronology of the echoes of the packets received, extracted at the end of the experiment via Wi-Fi.

We have chosen to configure all our nodes in 'Client' mode. Other modes exist (Relay, Router, etc.) and modify the priority and nature of background traffic, but the 'all clients' option simplifies deployment and is a pragmatic approach in the field. Client nodes are directly accessible from a terminal (smartphone, computer) and require no advanced configuration or specific strategic placement, other than judicious positioning in accordance with standard military transmission rules (high point, line of sight, distance from electromagnetic sources, etc.).

This module records packets as follows :

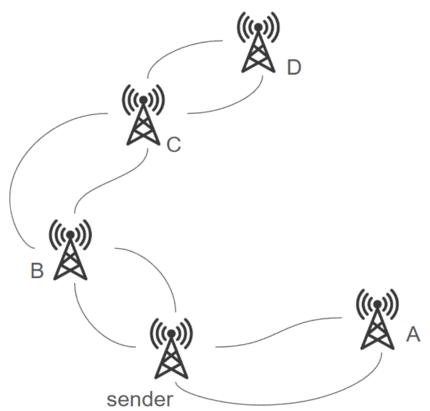


Figure 10: Exemple of mesh topology

When the sender broadcasts a packet :

- **sender-A** : Packet logged by A, A broadcasts an echo, Echo logged by sender
- **sender-B** : Packet logged by B, B broadcasts an echo, Echo logged by sender
- **sender-C** : Packet logged by B, B broadcasts an echo, this echo is logged by C, C broadcasts an echo, this new echo is logged by B, B broadcasts an echo, logged by sender
- **sender-D** : Packet logged by B, B broadcasts an echo, this echo is logged by C, C broadcasts an echo, this new echo is logged by D, D broadcasts an echo, this echo is logged by C, C broadcasts an echo, this echo is logged by B, B broadcasts an echo, logged by sender

This sequence of rebounds has a limit defined by the number of hops, which is decremented with each rebound. The packet contains two integers that count the number of hops performed and the remaining limit. The packet carries other information such as a unique packet identifier (defined by the sender, immutable). When recording the communication, the receiving node records the name of the sending and receiving nodes (these names change with each hop), its position as well as the position of the sender (known thanks to Meshtastic position and NodeInfo packets).

Python scripts for data analysis : A Python-based analysis pipeline was developed using established libraries (e.g., pandas/NumPy for tabular processing, GeoPandas for spatial operations, and matplotlib/seaborn for visualization). The scripts operate on datasets that are prepared in advance and cleaned of errors; data cleaning is not performed autonomously by the code. Sorting, filtering, and association rules—defined for this study (such as node-ID aggregation,routes reconstruction, and per-hop aggregation)—are then applied to derive indicators aligned with the study's questions. Finally, results are rendered as tables, maps, and charts configured according to the analysis needs.

3.2 Experimental setups

The experimental protocol is based on 3 major phases :

- Coverage and range : it consists in peer-to-peer measurements with different radio settings, majorly around buildings and at human height.
- Principal experiments : the deployment of a mesh network between the Passerelle and the Grande Bosse with the double objective to determine the maximum range between two high points and to evaluate the performances of the mesh and feasible ranges over multiple hops.
- Tactical scenario : The goal is to find out if it is reliable and relevant, the usability of such networks and the characterisation of the traffic.

3.2.1 Coverage and range

First experiment :

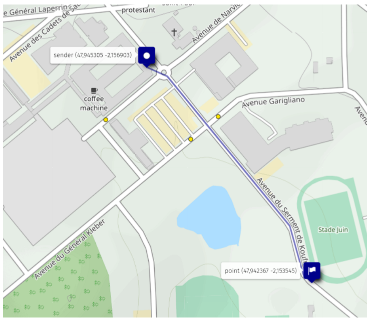
Objective : Provide a convenient way to determine the order of magnitude of radio quality between nodes for LoRa modules running Meshtastic in variable environments, and obtain intuitive results for LoRa configuration:

Table 3: configuration

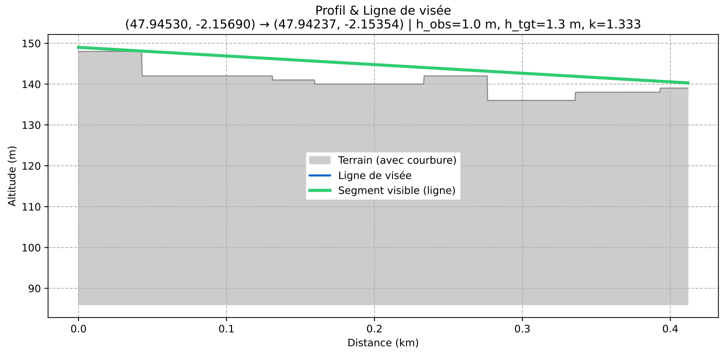
Bluetooth	**Wifi**	**Spreading factor**	**TX power**	**LoRa region**
Off	Off	Long-Fast	1dBm	EU 868

We conducted a LOS range test using two Meshtastic LoRa nodes using the above configuration. Nodes were separated progressively. The moving node transmitted 1 packet each 10s for 330 s (33 packets). Using the serial port, the listening node logged each packet RSSI and SNR.

We define maximum LOS range as the largest separation without any loss. Beyond this distance we logged a transmission failure leading to the end of the experiment. This method allowed us to reliably record the practical RSSI and SNR thresholds that can be sustained by the Meshtastic solution.



(a) Situation map



(b) Line of sight profile

Figure 11: Experimental setup and topographic analysis

Second experiment :

Objective : Determine the practical range of a LoRa module in an urban environment. Identify the modules' ability to overcome the presence of masks.

Table 4: configuration

Bluetooth	Wifi	Spreading factor	TX power	LoRa region
Off	Off	Long-Fast	22dBm	EU 868

To do so, we chose the following experimental protocol : The reciever was fixed somewhere while the sender was moved around to put to the test the connection capabilities of the devices. The *camp bâti* was ideal for this test because of the buildings which represented masks for the signals. During this experiment, the packets would contain data such as the SNR and RSSI to allow a packet-to-packet follow of the evolution of the quality of the signal. This experiment was made at maximum power of emission ; even though the european limits are set to 27dBm [6], we could not go higher than 22dBm which was the physical limit of the devices.

Figure 12: Position at packet transmission and radio quality (RSSI)

3.2.2 Performance logging for a mesh network

Objective : The objective of this field experiment is to deploy a Meshtastic-based LoRa mesh across representative terrains to (i) quantify end-to-end latency under controlled traffic and varying hop counts, and (ii) determine the maximum communication range (elongation) achievable with and without relays. Specifically, we will measure, latency and signal quality distributions, versus hops and distance. Secondary goals include isolating the impact of environmental conditions (line-of-sight, foliage, urban obstruction) and producing guidelines for node placement that minimize delay while maximizing coverage.

Table 5: inventory

Model	Frequency	WIFI	BLE	GNSS	QUANTITY
HELTEC-V3	868MHz	OFF	OFF	FIXED	2
TBEAM	868Mhz	OFF	ON	ON	2

The positioning of the nodes is designed to simultaneously optimise (i) the line of sight between critical relays, (ii) link redundancy in order to tolerate node failure, and (iii) service continuity in valleys where diffraction and terrain masks degrade the direct link. Two complementary approaches might be used:

- Planning guided by topographical analysis (DTM/DEM, elevation profiles, mask and Fresnel zone maps) to select high points and rebounds ensuring at least one alternative route per segment.
- An empirical 'stretched fabric' method consisting of iteratively deploying nodes at the measured range limit, then inserting additional relays until the target PDR and latency criteria are met in enclosed areas.

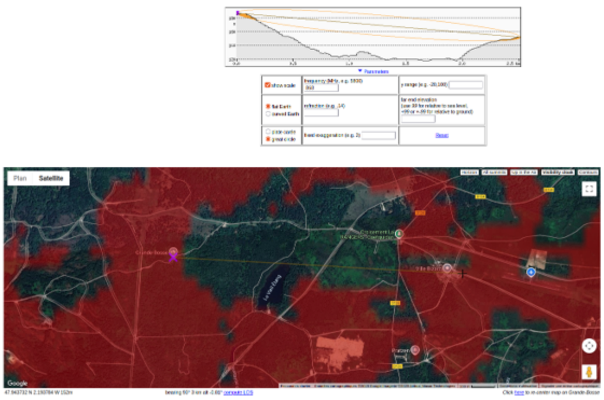


Figure 13: Line of sight and possible coverage in working area

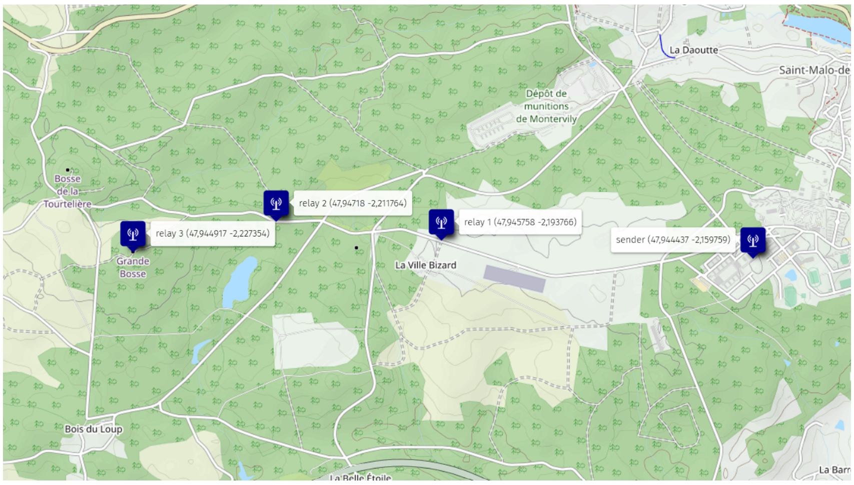


Figure 14: Situation map of the mesh network

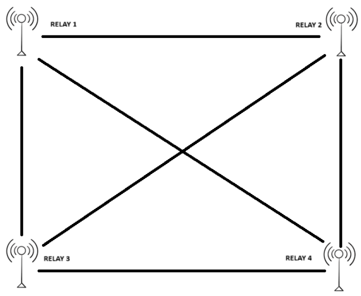


Figure 15: Mesh structure for the experience

It must be understood that the network was, during the experience, as described on Figure 14 ; it is indeed a mesh network that has been stretched.

Phasing and conduct of the experiment Fixed positions were assigned to the HELTECH units via configuration, whereas the single T-BEAM unit was configured to update its position periodically using the Meshtastic GPS update mechanism. The Sender node was installed on the *Passerelle* roof with a 4 dBi antenna, a tripod, and a portable battery. All remaining relays were then positioned iteratively using the previously described stretched-fabric method.

CSV logs gathering : The CSV log files generated by the modified RangeTest module are stored in the ./static directory of the ESP32s. They are retrieved via Meshtastic's embedded web server, which allows resources in ./static to be downloaded. At this stage, collection remains tedious and highlights the differences in functionality between the mobile application, the Python CLI tool, and the web configuration interface.

The procedure adopted consists of:

- Configure a local Wi-Fi network (mobile access point, Wi-Fi router, etc.) whose settings have been previously saved on the devices.
- Reactivate Wi-Fi at the end of the experiment (it is disabled during measurements so as not to interfere with RSSI and SNR readings).

- Use a Python script to access the nodes' web servers, then extract the CSV files via curl.

The raw CSV files are then ready for the cleaning and repair phase.

10.89.115.80_log.csv	07/10/2025 19:45	Fichier source Co...	17 Ko
10.89.115.86_log.csv	07/10/2025 19:49	Fichier source Co...	42 Ko
10.89.115.152_log.csv	07/10/2025 19:48	Fichier source Co...	83 Ko
10.89.115.234_log.csv	07/10/2025 19:46	Fichier source Co...	30 Ko

Figure 16: raw csv files from curl requests

Data cleaning and repair Temporal demarcation of the campaign : identify, in each log, the time interval matching with the targeted experimental phase (the files can contain remaining data from former experiences). Restoration of GPS position in particular : the initialisation delay of GPS receptors can produce invalid coordinates (for example 0;0). These anomalies are detected and corrected.

```
00:00:54,7,2956754304,1130078724,,0.0,0.0,0.0000000,0.0000000,0,367589,54645,4294654352,6.00,-73,3,0,0,0.0,"BENCHMARK|7|367589|2956754304|3|0"
00:01:54,8,2956754304,1130078724,,0.0,0.0,0.0000000,0.0000000,0,427742,114497,4294654051,6.00,-80,3,0,0,0.0,"BENCHMARK|8|427742|2956754304|3|0"
00:03:54,10,2956754304,1130078724,,47.9444643,-2.1597975,47.9443305,-2.1691940,159,548049,235241,4294654488,4.25,-94,3,0,0,699.50,"BENCHMARK|10|548049|2956754304|3|0"
00:04:55,11,2956754304,1130078724,,47.9444643,-2.1597975,47.9442820,-2.1695583,157,608203,295486,4294654579,0.25,-102,3,0,0,726.74,"BENCHMARK|11|608203|2956754304|3|0"
00:05:55,12,2956754304,1130078724,,47.9444643,-2.1597975,47.9442813,-2.1695521,158,668356,355637,4294654577,-7.50,-110,3,0,0,726.28,"BENCHMARK|12|668356|2956754304|3|0"
```

Figure 17: example of invalid data

Pre-processing and selection of variables After the repair, an initial process suppresses irrelevant information for the analyse :

- Payload and `sender_name` (debugging field)
- `distance_m` is replaced by a new column distance recalculated from the corrected coordinates.

The columns `sender_node` and `receiver_node` contain long IDs in full format. In reality, they represent a hexadecimal interpretation of the node's name : for example, 2956754304 is node B03C7D80's real name, short name 7D80. The time stamp column is not used because the nodes' clocks are not synchronised. The exploited emission and reception times are `send_time` and `recieved_time`. This lack of synchronisation can lead to unreasonable durations (like 50 days for example) during calculations ; these values are replaced by NaN so they won't be analysed. In the end, the exploitable file contains 12 columns.

```
23,a204,7d80,1330048,1334780,-8.0,-104,3,0,1,1435.0,4.732
24,7d80,a444,NaN,NaN,-0.75,-103,3,0,0,1448.0,NaN
24,7d80,a204,NaN,NaN,-3.75,-105,3,0,0,1437.0,NaN
24,a444,7d80,1390202,1403744,2.75,-103,3,0,1,1435.0,13.542
24,a204,7d80,1390202,1395854,-2.75,-108,3,0,1,1435.0,5.652
25,a204,7d80,1450356,1456247,3.0,-103,3,0,1,1435.0,5.891
25,7d80,a444,NaN,NaN,0.0,-101,3,0,0,1463.0,NaN
25,7d80,a204,NaN,NaN,-0.75,-102,3,0,0,1437.0,NaN
25,a444,7d80,1450356,1463521,-9.5,-104,3,0,1,1435.0,13.165
26,a444,7d80,1510510,1515623,2.0,-104,3,0,1,1435.0,5.113
26,a204,7d80,1510510,1521206,-2.25,-110,3,0,1,1435.0,10.696
26,7d80,a444,NaN,NaN,-1.75,-103,3,0,0,1449.0,NaN
26,7d80,a204,NaN,NaN,-5.25,-107,3,0,0,1441.0,NaN
27,7d80,a204,NaN,NaN,5.5,-78,2,1,0,1543.0,NaN
27,7d80,a444,NaN,NaN,-1.75,-103,3,0,0,1440.0,NaN
27,a444,7d80,1570664,1578029,1.5,-104,3,0,1,1435.0,7.365
27,a204,7d80,1570664,1605209,-3.0,-110,2,1,1,1543.0,34.545
28,a204,7d80,1630818,1637180,-2.0,-109,3,0,1,1750.0,6.362
```

Figure 18: extract of the exploitable final file

433Mhz parallelisation : Attempted parallel trials in the 433 MHz band were unsuccessful: even at maximum transmit power and with settings mirrored from the 868 MHz profile, end-to-end range did not exceed 100 m, which was unexpectedly poor. To investigate this anomaly, we measured the supplied 433 MHz T-Beam whip antennas with a vector network analyzer; the return-loss minimum (S11) was near 375 MHz indicating severe impedance mismatch. This out-of-band tuning explains the short range and prevented a meaningful 433 MHz evaluation; accordingly, 433 MHz results are excluded here, and we attribute the failure to defective antennas rather than to Meshtastic or the LoRa PHY at 433 MHz.

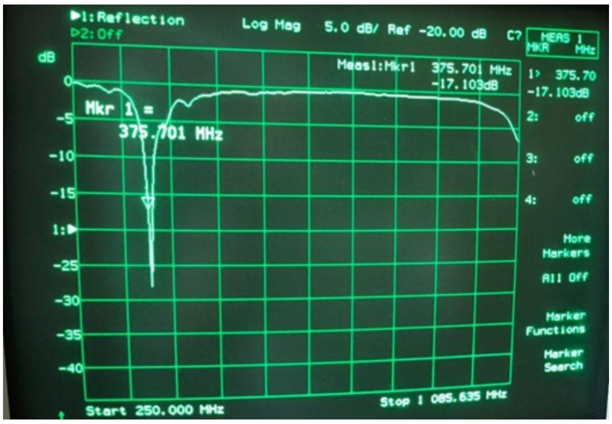


Figure 19: 433 MHz Antenna — VNA Measurement

3.2.3 COBRA operational scenario

Objectives : We conduct a concrete case study called *Mission COBRA*, to evaluate performance under a credible scenario, partition traffic into mission versus control components, compute packet delivery ratios, and measure how relaying mitigates performance loss during degradation events.

Context : Three commando teams have been tasked with identifying a potential enemy infiltration route in the urban area of the village of Coëtquidan. The team leaders—Fox 1, Fox 2 and Fox 3—arrived from the seaside resort "Bazar Beach" and must gather data and confirm the location before deploying a rapid reaction unit. With the village under enemy control, utmost discretion is paramount. A covert operative who previously recognised the site has installed a LoRa module equipped with a pax counter along the infiltration route. The three teams must therefore use this device to verify the threat for the command post.

Deployment Using Fig. 20 as a reference, Fox 1 issues the following orders: Fox 2 moves to waypoint X11, Fox 3 to waypoint X12, and Fox 1 to waypoint X13. This configuration ensures that Fox 3 maintains visual contact with the objective, while Fox 2, positioned at a higher elevation, serves as a radio relay for Fox 1. Fox 1 remains in rear support, maintaining communications with both elements.

Prior to the operation, an undercover agent infiltrated the building and discreetly installed a Heltec device equipped with the Paxcounter module. This setup enables real-time monitoring of the occupancy of a specific room, providing valuable intelligence on the presence and movement of individuals within the target area.

To enhance communication reliability in a dense urban environment with significant signal obstruction, Fox 3 activates a high-power relay equipped with a 4.4 dBi antenna. This setup improves link quality and coverage despite the presence of numerous physical barriers.

Table 6: inventory

Model	**Frequency**	**WIFI**	**BLE**	**GNSS**	**QUANTITY**
HELTEC-V3	868MHz	OFF	OFF	FIXED	2
TBEAM	868Mhz	OFF	ON	ON	3

We used a 4.4 dBi antenna mounted on tripod to act as a powerfull relay.

Table 7: configuration

Bluetooth	**Wifi**	**Spreading factor**	**TX power**	**LoRa region**
Off	Off	Long-Fast	22dBm	EU 868

Organisation

- Fox 1 (commando team leader): 1 LoRa module (communications)
- Fox 2 (relay): 1 LoRa module (communications)
- Fox 3 (forward element): 1 LoRa module (communications)

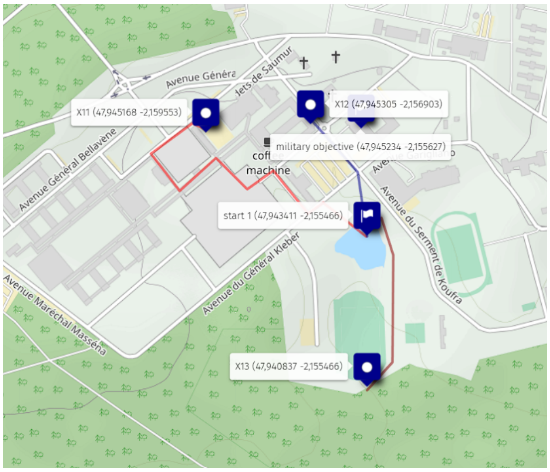
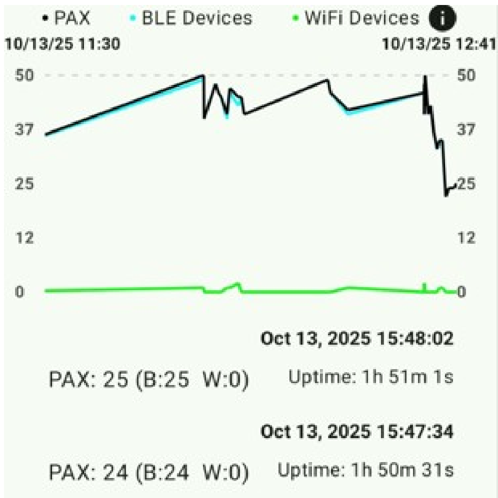


Figure 20: The deployment of the 3 groups

Paxcounter : As noted earlier, the undercover agent deployed a Heltec node running Meshtastic with the Paxcounter module enabled. In Meshtastic, Paxcounter is a sensor module that estimates crowd density by passively scanning for nearby Wi-Fi probe/beacon frames and Bluetooth Low Energy advertising packets. It does not associate with devices or collect persistent identifiers; it reports only aggregated counts (e.g., Wi-Fi count, BLE count) at configurable intervals, forwarded as encrypted telemetry frames over the mesh. Meshtastic Android app enables live-tracking of the paxcounter data (see Figure 21b).

(a) Heltec module running paxcounter



(b) Live extracts from paxcounter data

Figure 21: Paxcounter sensor node

4 Results

4.1 Range tests

The first experiment, deliberately simple, allowed us to obtain a small data set to determine an order of magnitude of $-100dBm$. (As shown on Figure 22).
Although the average RSSI drops rapidly and reaches very low levels (less than -80 dBm), we have not yet reached the maximum limit, and thus, because of our strict PDR choice. Empirically, the edge-of-link measurements clustered around $RSSI \approx -120$ dBm with $SNR \approx -20$ dB. These values should be read as orders of magnitude rather than hard thresholds.

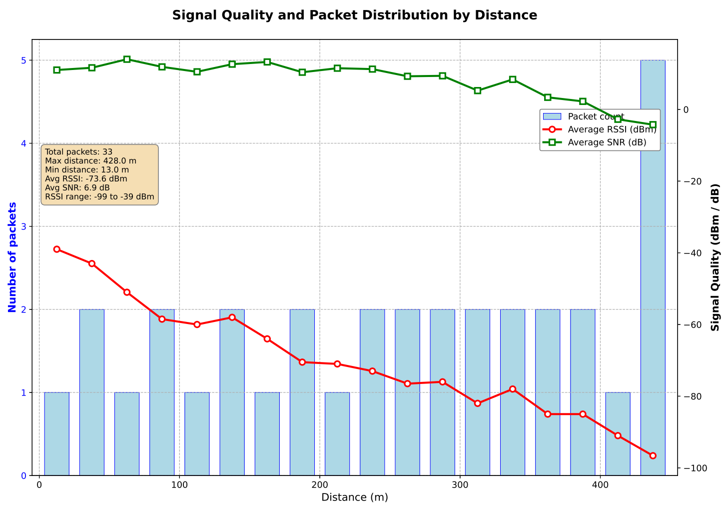


Figure 22: SNR/RSSI over distance — 1dBm Long Fast — LoS case

According to different radio settings (SF, BW, freq, power), using Meshtastic presets (Short fast, long moderate, long fast).

Long range test between high points: keep measurements between 7d80 and a444 once at the big bump and say that it confirms the simulation (the one done in the first session).

- Short-fast: effective but limited range, this is good for dense networks coupled with low TX PWR for lab tests.
- Long moderate: ToA too long and channel occupancy too high, congestion in message queues.
- Long fast: good compromise, very good range with more reasonable ToA, this is used for the rest at TX PWR max for maximum coverage (22 dBm on SX1262 chipsets)

Lower SFs favor frequent communications in dense meshes by minimizing airtime, whereas higher SFs are required to sustain links in stretched topologies with significant inter-node separation. Meshtastic currently enforces a single radio profile per channel, precluding mixed PHY parameters within the same mesh channel. The reliability of measurements is contingent on environmental factors, including antenna siting/orientation and the absence of GNSS indoors. While elevated placements considerably extend range, wind resistance must be ensured; the collapse of a high-ground relay was observed, after which the network maintained service through rerouting.

4.2 Mesh network performances evaluation

Exploratory analysis — correlation matrix : A correlation matrix (visualised in heatmap with Seaborn) gives a global view of linear associations between numeric variables. It summarises, for every peer, the Pearson coefficient between -1 and 1 : close to 1, positive and strong relation ; close to -1, close negative relation ; around 0, no remarkable linear relation. This global vision helps the formulation of hypotheses and prioritise analyses.

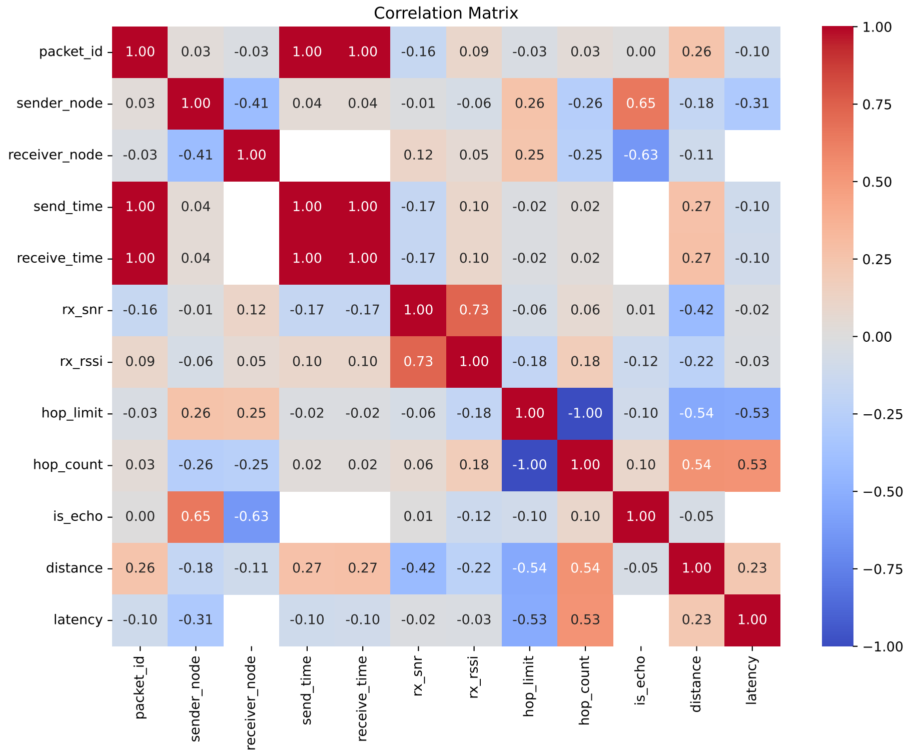


Figure 23: Pearson coefficient linear correlation matrix

Figure 23 shows a strong positive correlation between end-to-end latency and hop count. In mesh networks, latency typically accumulates per hop due to airtime, channel access, and per-node processing; this trend can become nonlinear under congestion or very low bandwidth. Over the distances tested, RSSI decreased and SNR degraded with separation, as evidenced by our range-test beacons; these measurements are consistent with LoRa links operating reliably up to approximately 5 km under favorable conditions. Within the observed SNR/RSSI range, latency did not measurably depend on signal quality because no retransmissions were triggered; latency was instead dominated by deterministic per-hop airtime and processing.

Protocol-linked interpretations : The observed correlation between `sender_node` and `is_echo` can be explained by the sequential powering up of the nodes, with time lags inducing different log volumes depending on the node.

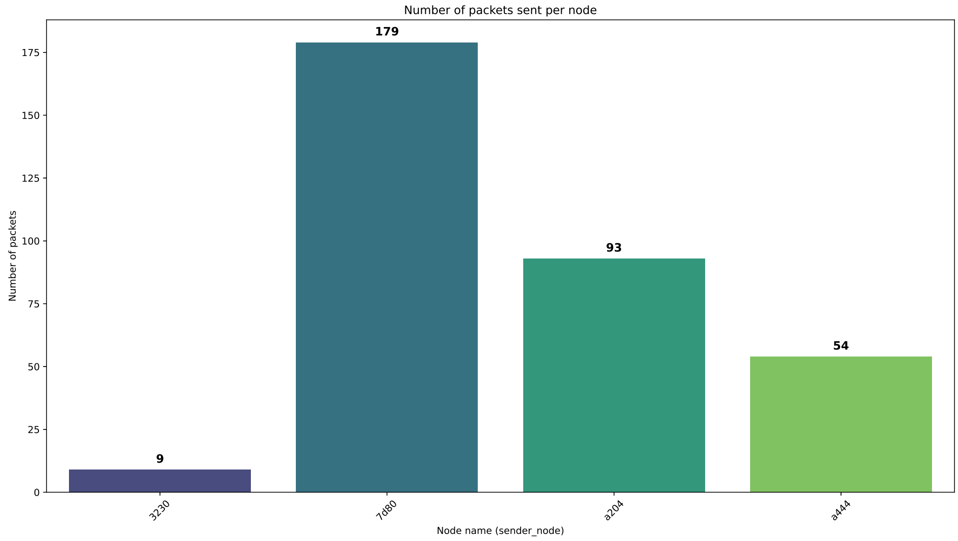


Figure 24: Packet volume per node

Node 3230 has a particularly low number of echoes: its precarious installation caused the antenna to disconnect, leaving it off-network most of the time.

Latency: beyond geometric distance The intuition that 'the further away, the slower' needs to be qualified. In a LoRa/Meshtastic network, radio travel time is negligible on a scale of kilometres.

No found correlation between latency and radio signal quality. It should be noted that the intuition that latency could increase as radio signal quality degrades is incorrect. In fact, a coefficient of -0.02/-0.03 for snr/rssi with latency indicates a clear lack of correlation between these two data points. This lack of correlation thus demonstrates the error correction performance of the Meshtastic protocol.

Significant correlation between the number of hops, distance and latency. Naturally, distance and number of hops are two fairly correlated factors, simply explained by the need to hop when the distance becomes critical. The link between latency and the number of hops can be explained by the very simple principle of Chinese whispers. Each retransmission by a node adds delay to the overall transmission, just as in oral transmissions between humans.
Latency results primarily from:

- airtime (depending on the spreading factor, bandwidth, coding rate and frame size),
- listening and backoff mechanisms (listen before transmit, anti-collision),
- queues in relay,
- possible retransmissions.

Using an empirical approach without attempting to create a precise model, we can approximate latency with an equation that incorporates our main latency factors.

$$\text{Latency}_{e2e} \approx \alpha + (h + r) \cdot (\text{airtime}_{\text{PHY}} + t_{\text{proc}} + t_{\text{access}})$$

- h: hop count
- r: number of retransmissions
- α: fixed constant (linear undermined factors)
- $airtime_{\mathrm{PHY}}$: Time of travel on physical level
- t_{proc} : processing time per node
- t_{access} : waiting time due to busy channel

The main determining factors are therefore the number of hops and the network load. Distance only plays an indirect role by imposing multi-hop routes in the presence of obstacles or by inducing retransmissions, which increases losses and delays. The following analyses will quantify these multi-factor effects on latency and signal quality.

Degradation of radio signal quality with distance. The increase in inter-node distance results in a significant decrease in the average and median RSSI, down to -117.5 dBm, a value close to the receiver sensitivity for our LoRa settings. Conversely, the RSSI appears to increase with the number of hops; this is a measurement artefact: our logging system associates the position of the initial transmitter but only records the RSSI of the last hop, which is typically shorter. This peculiarity skews multi-hop comparisons and does not reflect end-to-end quality. (Figure 25)
Under the test conditions described, the practical range limit is around 5 km. This estimate corresponds to the point where radio quality (RSSI/SNR) and end-to-end performance fall below the defined usage threshold. (-120 dBm for RSSI).

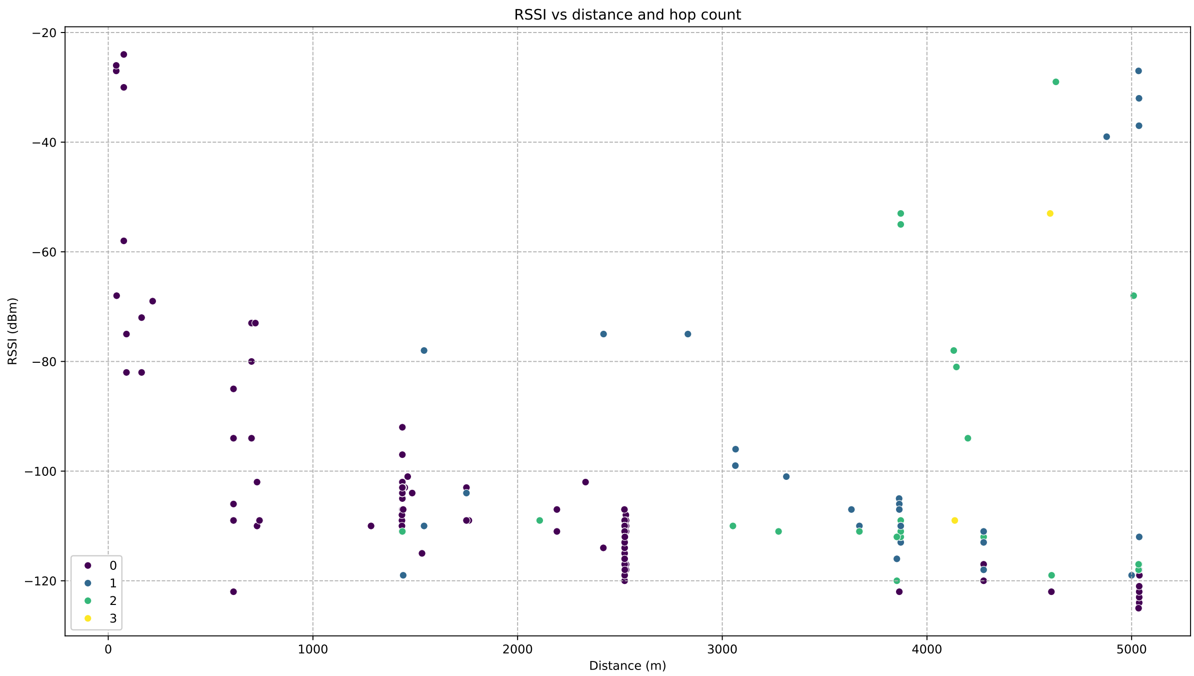


Figure 25: RSSI vs distance and hop count

To eliminate this measurement artefact, we analyze only single-hop links, ensuring that the recorded RSSI corresponds to the same transmitter–receiver pair as the logged position.

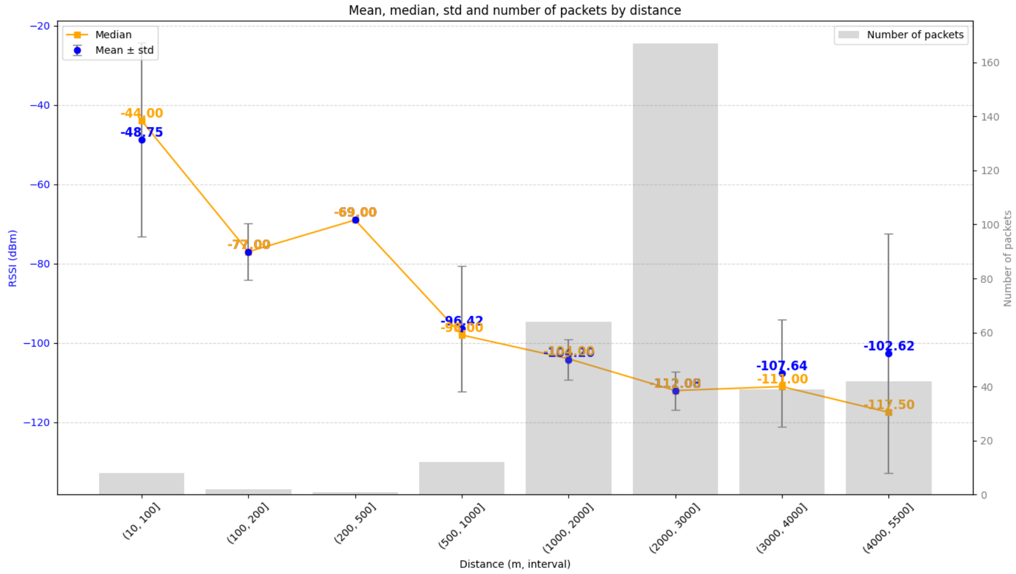


Figure 26: Mean, median, std and number of packets by distance

SNR depending on distance and number of hops: The vertical bars visible on the Figure 27can be explained by the fact that some of the nods did not move. At constant abscisse (distance/position linked to the static node), the movements of the other node vary the signal quality (RSSI/SNR), which explains the spreading of statements on the y-axis, whilst the vertical dispersion highlights the immediate fluctuations of the canal and of geometrical distances.

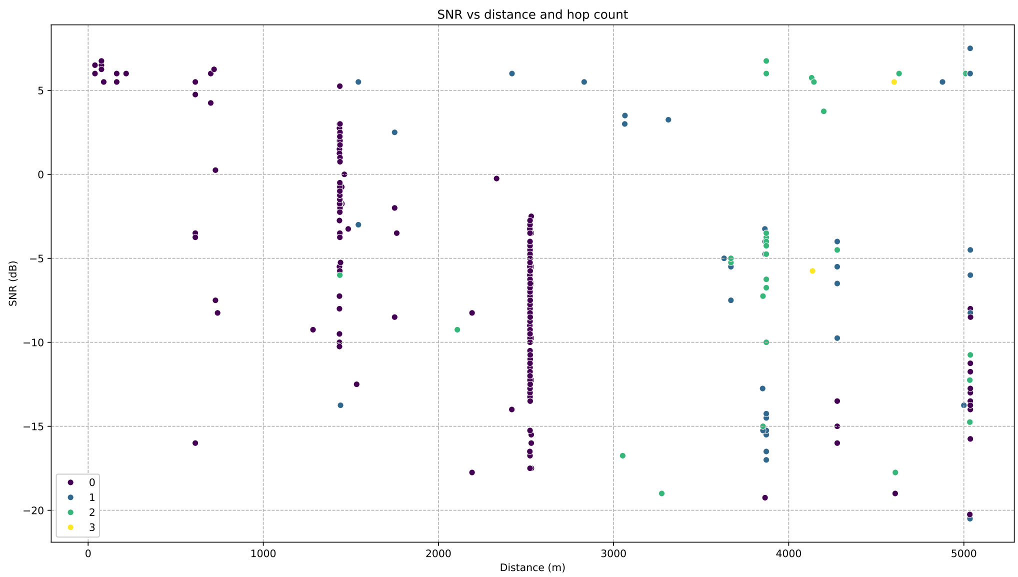
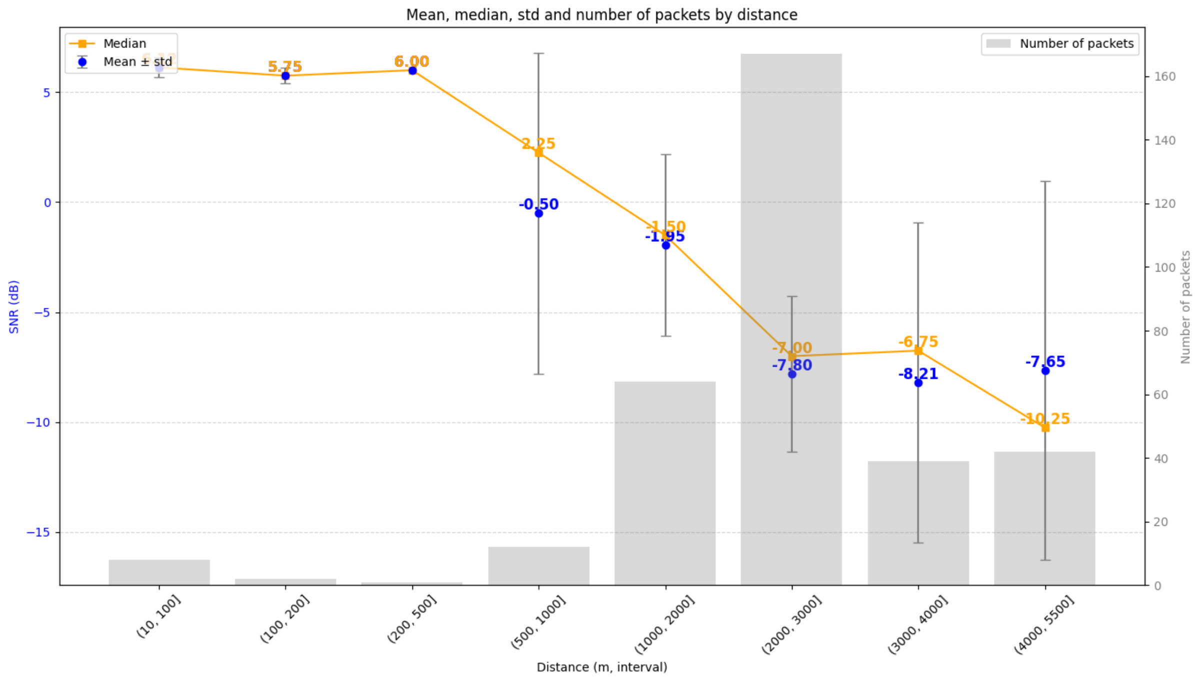


Figure 27: SNR vs distance and hop count

Following the same method, since the quality of a radio link is also assessed via the SNR, a statistical analysis is required to determine its distribution, central tendency and variability. These data validate the tests previously conducted with a maximum practical RSSI of -120 dBm and an SNR of -20 dB. Necessarily involving fluctuations due to the practical use of the mesh network./



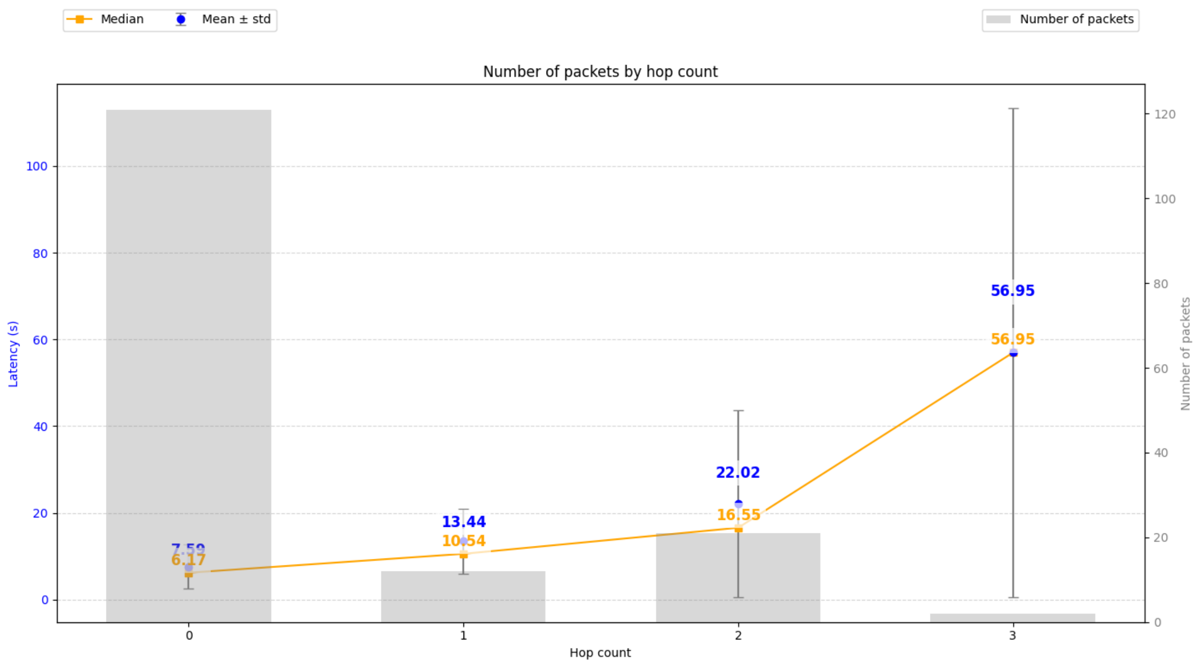


Figure 28: Number of packets by hop count

Evaluation of the latency depending on the number of hops: End-to-end latency increases significantly with the number of hops (Figure 28). In direct connection (0 hops), the average is 6.17 s, which sets the offset of the measurement chain. With 1 hop, the average latency reaches 10.54 s (+4.37 s), then 19.55 s with 2 hops (+9.01 s), suggesting a cost per hop that is not strictly constant but increases with load and channel access. At 3 hops, the average jumps to 56.95s ; we interpret this with caution because the sample size is small and the loss rate is high, which artificially lengthens the observed delays (retransmissions, queues, backoff). We will report the standard deviation and/or IQR for each condition and distinguish between 'successful' latencies and lost packets in order to avoid censorship bias; failing that, we will provide confidence intervals and explicitly mark the '3 hops' condition as unrepresentative.

4.3 Military usage assessment

Packet delivery ratio The hands-on case study allowed us to understand certain advantageous aspects of using LoRa in a tactical context.
First, it enables dynamic relay deployment. When Fox 1 was too far from Fox 2 and Fox 3 while moving towards X13, it was possible to deploy a covert mid-way LoRa relay to extend the signal range.

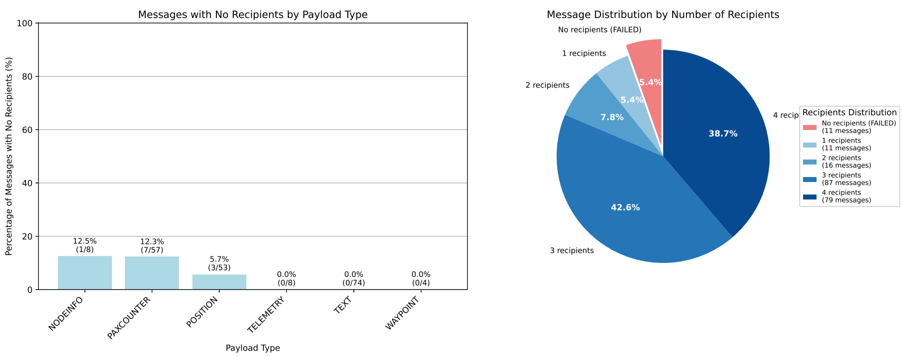


Figure 29: Delivery ratio of packets depending on payload type

Moreover, the received packet results showed the network's high reliability. Indeed, only 5.4% (Figure 29) of messages were not received. None of these dropped packets contained critical information because important data are prioritised. Datas like telemetry which are less important or technical messages which represent 18.5% (Figure 30/Figure 31) of margin of manoeuvre to accept potential losses.

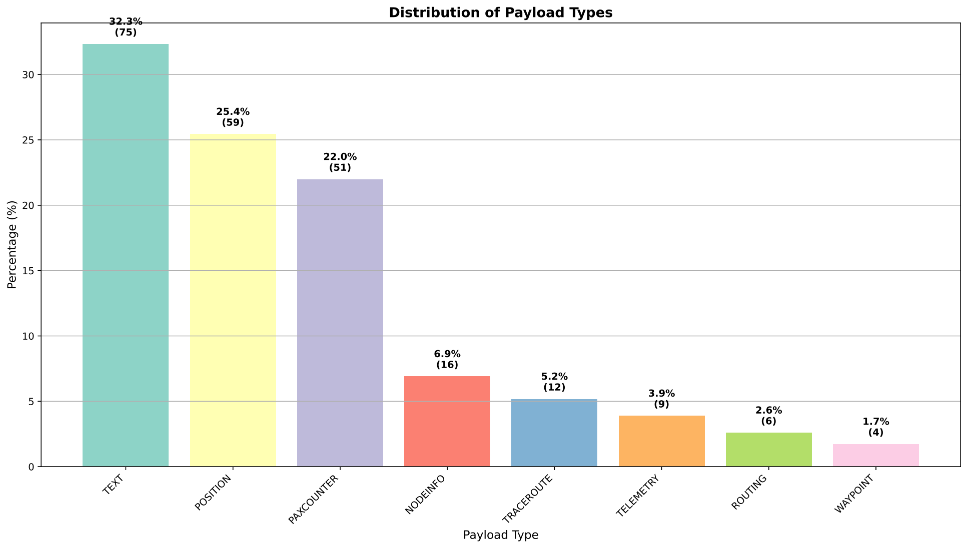


Figure 30: Proportion of payload type

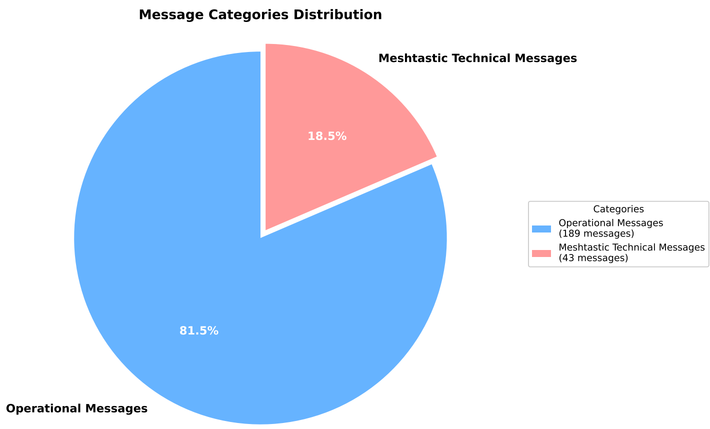


Figure 31: Operational vs technical traffic

Overall, communication within the group was successful and the mission proceeded flawlessly. The only drawback was the lack of hierarchy on the network: there was no differentiation between the group leader and the other members. This limitation stems directly from the Meshtastic software.

5 Discussion

5.1 Interpretation of the results

Strengths :

- the mesh protocol is efficient. In addition, the ecosystem and community surrounding Meshtastic are very powerful (client apps development for example). It is reliable with good performances. Finally, even though we did not have the opportunity to try it, an ATAK module is available, specifically targeting C2 and tactical awareness, which is a valuable asset.

Limits :

- First, radio settings are fixed, meaning if the parameters need to be changed, the module needs to be restarted ; the resulting problem is that it loses the synchronisation between the devices and it won't be automatically restored.
- The system is not flexible nor allows power adaptation. For example, on the following topology :
  group(6 devices) - relay ———- relay - group(6 devices) ;
- The relays would need not to be on the same emission power than the groups so to reach one another but all devices have to be on the same physical canal with the same parameters (except for the emission power which cannot be set dynamically).
- Meshtastic is not tolerant to connectivity disruptions. If nodes are not reachable when a message m is sent, then m is lost and cannot be received later. (The Store and Forward module is an attempt to address this issue but not sufficient/efficient).

We came across many problems while deploying the network :

- The retro-engineering process is long and time-consuming ; the project is consequent (it is composed of more than 700 files and 110.000 lines of code) and the appropriation process was not granted.
- A timestamp problem while sending packets. The devices lose their timestamp synchronisation every time they are rebooted, which complicates the synchronisation of all devices. To solve this problem, we chose not to use time stamps as a reference but to adapt the logs with special messages to identify the beginning and end of the experiments. In addition, the analysis of the logs was not available immediately so we followed the packets received by the sender to monitor the network.

Strenghts and weaknesses of Meshtastic With some modifications on the system and a few tests, Meshtastic revealed itself as a powerful and reliable structure to use in tactical situations. Fluid, not that much latency, intuitive, accessible online, versatile... It has many advantages. However, the different experiments revealed weaknesses **Connecting two points spaced 5 kilometers apart in direct line of sight proved successful for us.** Indeed, it demonstrated LoRa's transmission capabilities at low power (22 dBm), and allowed us to assess the effectiveness of relays which depends heavily on their placement. The higher a relay is positioned and the clearer its line of sight, the more efficient it becomes.

Unfortunately, the experiment also revealed some weaknesses of these modules: one unit fell and its battery unplugged, shutting the module down. This highlights an important point : low-cost devices are fragile and require appropriate physical protection when deployed on the field.

5.2 Potential applications in military and civilian contexts

Original civilian use : When connected to the IoT, it enables sensor data collection in agriculture, healthcare and fire-safety applications. However, its low energy consumption, minimal cost and operation on a civilian frequency can all serve as advantages in a military context.
Firstly, its low cost allows for mass deployment. For example, LoRa nodes can be emplaced as relays on a mountain ridge without concern for retrieval. It is also conceivable to air-drop large numbers of LoRa devices over an area to facilitate enemy detection or to establish communication relays.
Moreover, even though there is no physical security, Meshtastic channels can be encrypted with AES keys. This built-in encryption can therefore help mitigate cybersecurity risks.

The compact size of LoRa makes it suitable for infiltration and surveillance missions when used with a Pax counter.

Figure 32: Radio hidden inside a book

6 Conclusion

6.1 Advantages

Experiments have shown that LoRa can transmit and receive data with low RSSI and SNR. Even when signal quality is poor, LoRa is able to recover the transmitted packet and distinguish it from ambient noise, which could indicate some kind of resilience to jamming.

Moreover, the vast majority of packets reach their destination and the error rate remains low, highlighting the network's high reliability—ideal for tactical communications where positions updates and orders need to reach their destinations.

We can also highlight the resilience of the LoRa network, as demonstrated in the experiment linking the roof of the headquarters building to the *Grande bosse*. During this test, a node attached to a branch broke off; however, the network kept operating without that node, and the two ends of the network stayed connected using other routes. Thus, communication can be maintained during a conflict even if a LoRa node is destroyed.

Beyond the pax counter module based on an off-the-shelf ESP32 library, the radios can be fitted with additional sensors by employing an endless variety of modules, such as jamming detectors like the RFeye Scanner or LabSat ION.

6.2 Problems

Meshtastic is an open-source project still under development. It is therefore not fully mature and evolves constantly, which can complicate its use. For example, access to the flash filesystem is difficult (no access via the serial port), and synchronizing radio timestamps becomes unfeasible due to latency.

On the hardware side, the ESP32 boards that we used remain fragile and false contacts occur frequently. During our experiments, two of our devices broke before protection cases were built.

The latency measured in our experiments can be a limiting factor in a conflict if the network density increases. Even though such limit was not reached, we could expect traffic congestion if fifty nodes were to communicate on the same area. For more dense networks, radio settings must be adapted, especially the spreading factor.

Moreover, having all the nodes communicate on the same radio channel (i.e., same spreading factor, bandwidth and coding rate) prevent nodes to dynamically adapt to their neighbourhood. For instance, if a node that is part of a dense subgroup (short links, low SF, low power) must also communicate with a distant relay (long link, high SF, high power), it cannot switch from one setting to another, and thus cannot optimize both.

Regarding Meshtastic, there is a lack of security at the physical layer of the nodes, which poses a security risk if this firmware is used for military applications. This risk can however be mitigated by blending meshtastic military traffic into other LoRa traffic like LoRaWAN communications in a smart city.

6.3 Possible Solutions

Add a physical security layer to the radios: encrypt the LoRa radio data or even deploy a “kill virus” in the device should the enemy

6.4 Possible Solutions

Add a physical security layer to the radios: encrypt the LoRa radio data or even deploy a “kill virus” in the device should the enemy connect it to their computer.

In this vein, it would be ideal to develop a bespoke application to exploit LoRa technology in a military context. This application would manage packet handling and automatically select radio parameters based on the operational environment.

The hardware portion can also be reinforced for security with anti-tamper resin. For physical robustness, it would have been preferable to mould the enclosures entirely in resin and, in certain cases, to employ tropicalisation when the modules may be subjected to frequent impacts.

6.5 Key takeaways

The number of modules was insufficient for large-scale testing in a tactical mission and for practical field deployment. The limited time 80 hours spread over six weeks proved to be a constraining factor for this type of test. The lack of time also limited the number of results obtained at the conclusion of the experiments. For example, each experiment would have needed to be repeated several times to confirm the results and minimise sources of error. Cause of this lack of time we did not perform any theoretical simulations of SNR or RSSI as a function of distance for the LoRa radio.

Furthermore, while we intended to compare transmissions on 433 MHz and 868 MHz, particularly in forested environments, issues with the poor quality of the 433 MHz antennas (the antennas were defective) prevented us from carrying out experiments at that frequency, which drastically reduced the number of results.

Obtaining radio testing authorizations within the academy was difficult, and a lack of communication among the leadership led to several complications.

Nevertheless, the project enabled us to explore multiple domains, from computer science (networking protocols, embedded firmware development and data processing with Python code) to electronics (battery, antenna evaluation, signal modulation) or computer-assisted design (3D printing). The affordability of LoRa technology and its open-source applications also piqued our interest. Projects like this demonstrate that military technology can be tackled by anyone with scientific skills, regardless of financial resources.

Beyond mere intellectual accessibility, the low cost of these radios would allow them to contribute to a war effort if LoRa technology were deployed on a large scale. Indeed, similar to the low-cost drone manufacturing in Ukraine, it would be possible to produce compact radio modules for field delivery. Once configured, these radios would be ready for immediate use.

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